

DISCRIMINATION IN MULTI-AGENT SYSTEMS

Anonymous authors

Paper under double-blind review

ABSTRACT

We study how discrimination can emerge in a population of interacting learning agents. Each agent is modeled by a perceptron that maps a shared representation of an issue to an opinion, together with estimates of the reliability of other agents. The perceptron represents an effective opinion readout and can be viewed as a linear decision layer acting on representations produced by a large language model. Agents update both beliefs using an online Bayesian learning rule derived for a student learning from a teacher through a noisy channel. A large language model using in-context learning reproduces the qualitative form of both updates predicted by this rule, including the one that decides the direction of opinion change. Without group bias, the population spontaneously separates into two mutually distrustful camps. We then introduce an irrelevant group label. A fraction f_p of agents shifts its perceived agreement with a message by a strength p according to whether the sender belongs to the same group. Simulations over (p, f_p) produce four collective states: neutral polarization; discrimination in both trust and opinion; discrimination in trust but not opinion; and a frustrated state when agents favor the out-group. The outcome depends on both the strength and the prevalence of the bias. Individual measurements of bias are therefore insufficient unless they are aggregated at the population level. We define order parameters based only on pairwise opinion alignment and trust, so they can also be measured in systems whose internal representations are unavailable. The model therefore provides a simple framework for studying discrimination as a collective property of interacting learning agents.

1 INTRODUCTION

Learned agents increasingly interact in groups: they debate, review one another, negotiate, and simulate populations (Park et al., 2023; Du et al., 2024; Chuang et al., 2024). These interactions can produce collective behavior that is not apparent from any agent in isolation. We ask whether they can also produce discrimination: persistent differences in trust and opinion organized by a label that contains no information.

We isolate the role of interaction by using a model with an explicit learning rule. The model contains N perceptron agents and an agenda of P issues embedded in K dimensions. Each agent holds a Gaussian belief over its own weights and a separate estimate of every other agent’s reliability. After receiving an opinion, it updates both estimates using an online Bayesian rule projected back to a Gaussian distribution (Oppen, 1996; Caticha, 2020). The inferred channel noise is interpreted as distrust (Alves & Caticha, 2016).

We introduce group bias through a minimal representational error. A fraction f_p of agents shifts its perceived agreement with a message by $+p$ for an in-group sender and $-p$ for an out-group sender. The labels do not affect the issues or the initial conditions, and both groups follow the same rule. Any group asymmetry must therefore emerge from the interactions.

The simulations produce four states over the (p, f_p) plane. With weak or rare bias, the population polarizes independently of the label. With sufficiently common in-group bias, trust follows the label, and opinion may or may not follow it. When the bias favors the out-group, incompatible trust relations produce a frustrated state. The transition depends on both p and f_p : moderate bias has little effect when it is rare but becomes strong when shared by many agents. This means that per-agent measurements must be combined with population-level measurements to identify the collective state.

The mechanism acts through trust. Perceived agreement determines whether an agent increases or decreases its trust in a source, so a group-dependent error in agreement becomes a group-dependent trust update. The effect is not due to a failure of the single-agent inference rule. It appears when a rule derived for one student and one teacher is used throughout a population.

Contributions.

- We show that a large language model with frozen weights reproduces both mean-update functions of the Bayesian rule. For the opinion update F_w , a binary readout over 550 conditions has rank correlation $r_s = 0.90$ with the prediction and the same sign in 94.7% of cases; a distrusted source moves the model away from its statement. For the trust update F_μ , elicited reliability over 712 conditions has correlation $r = 0.85$ and 86% sign agreement (§3.2, Appendix E).
- We show that an irrelevant group feature can reorganize a population into four collective states (§3.3, §4.2). Agenda complexity, $\alpha = P/K$, determines whether discrimination appears in trust alone or in both trust and opinion.
- We define population-level diagnostics using only pairwise opinion alignment and trust (§4.2, §5). They do not require access to weights, gradients, or embeddings.

Relation to prior work. The model, its learning rule, and the two balance order parameters are due to Alves & Caticha (2021), as is the observation that the order of polarization reverses with the size of the agenda; §3.1 and §3.1 recap what the rest of the paper needs, and Appendix G gives the longer genealogy. New here are the substrate test of the modulation functions, the conservation law that explains the reversal (Appendix A), and everything concerning discrimination: the field, the phase diagram, the class order parameters, and what they imply for evaluation.

The model is not intended as a model of human societies. It establishes a narrower result for learning systems: a locally well-founded inference rule can turn an irrelevant label into stable group distrust when used by many interacting agents.

2 RELATED WORK

Societies of language-model agents. Multi-agent language systems have been used for population simulation (Park et al., 2022; 2023), factual debate (Du et al., 2024), opinion exchange (Chuang et al., 2024), large-scale coordination (De Marzo et al., 2024), and survey simulation (Argyle et al., 2023; Aher et al., 2023). Recent work studies collective behavior directly, including emergent conventions and bias (Ashery et al., 2025), systematic biases in debate (Taubenfeld et al., 2024), and organizational failures (Cemri et al., 2025).

Closest to our setting, El et al. (2026) measures more than ten thousand communities of language agents and fits a statistical-mechanics model with regimes of indifference, polarization, and consensus. We instead begin with a derived interaction rule and compute the population it produces. This makes it possible to study how an irrelevant group label changes the phase structure.

Bias and fairness in machine learning. The dominant framing locates group bias in data and in single-model decisions, as representational harm (Blodgett et al., 2020; Bender et al., 2021) or as allocative unfairness with formal criteria attached (Dwork et al., 2012; Hardt et al., 2016; Barocas et al., 2023). Our mechanism is complementary: an irrelevant feature affects the learning rule of multiple agents and changes their collective state. Detecting it therefore requires aggregating measurements across the population.

Opinion dynamics and social balance. Models of consensus and polarization in interacting populations run from bounded-confidence and mixing rules (Deffuant et al., 2000; Hegselmann & Krause, 2002) through culture dissemination (Axelrod, 1997) to reinforcement-driven echo chambers (Bauermann et al., 2020); see Castellano et al. (2009) for a survey. Representing agents as classifiers over an issue space (Caticha & Vicente, 2011) allows partial rather than binary agreement. We also use social balance over triples (Heider, 1946; Cartwright & Harary, 1956; Antal et al., 2005; Marvel et al., 2011), to distinguish two organized camps from a frustrated population. In our model, both opinion and trust relations are generated by inference rather than specified directly.

Optimal on-line learning, and affective polarization. The learning rule is the optimized on-line algorithm for the student–teacher problem, given a Bayesian reading by Oppen (1996) and an entropic-dynamics formulation by Caticha (2020); the identification of the inferred channel-noise level with distrust, and its promotion to a dynamical variable, is due to Alves & Caticha (2016). Appendix G gives the fuller genealogy. We study what this algorithm does at the population level after adding one class-correlated feature. The model also relates to the debate over whether groups dislike one another because they disagree or disagree because they dislike one another (Fiorina & Abrams, 2008; Iyengar et al., 2012; Iyengar & Westwood, 2015); we do not adjudicate this empirical question, but identify a model parameter that selects the order of the two effects.

3 LEARNING AGENTS

We model each agent’s opinion readout as a perceptron that learns from other agents. We first define its state and update, then compare the trust update with in-context learning in a language model, and finally introduce the group-dependent error.

3.1 AGENT STATE AND LEARNING DYNAMICS

A society of N agents discusses an agenda of P issues. Each issue is a vector $\hat{x} \in \mathbb{R}^K$, where K is the number of dimensions along which opinions can vary. All agents use the same issue representation but may assign it different labels.

Each agent carries a single-layer perceptron with weights $w_I \in \mathbb{R}^K$ and states the opinion $\sigma_I = \text{sign}(w_I \cdot \hat{x}) \in \{\pm 1\}$. The perceptron models an effective opinion readout rather than the full agent. It can be interpreted as a linear decision layer acting on issue representations produced by a large language model. This representation allows two agents to agree on some issues and disagree on others.

An agent’s state has two components. The *opinion* component is a Gaussian belief over its own weights, with mean $\hat{w}_I$ and covariance C_I . The *trust* component is a Gaussian belief, for every other agent J , over how unreliable J is as a source: mean $\mu_{J|I}$ (the *distrust* I assigns to J) and variance $V_{J|I}$. The joint belief is

$$Q(w_I, z_I \mid \hat{w}_I, C_I, \mu_{J|I}, V_{J|I}) = \mathcal{N}(\hat{w}_I, C_I) \prod_{J \neq I} \mathcal{N}(\mu_{J|I}, V_{J|I}), \quad (1)$$

The microscopic variables are $\hat{w}_I$, C_I , $\mu_{J|I}$, and $V_{J|I}$. The means determine opinions and trust, while the variances determine uncertainty.

Dynamics. At each step, an issue $\hat{x}$, an emitter e , and a receiver r are sampled uniformly. The emitter states σ_e , which the receiver treats as a noisy report of an unknown true label. The receiver’s estimate of the noise is its distrust of the emitter. A Bayesian update takes the belief outside the Gaussian family, so we project it back by matching the first two moments. This gives a closed online update. Appendix F gives the derivation.

Two scaled fields carry everything the update depends on:

$$h_w = \frac{\hat{w}_r \cdot \hat{x} \sigma_e}{\gamma_C}, \quad h_\mu = \frac{\mu_{e|r}}{\gamma_V}, \quad \gamma_C = \sqrt{1 + \hat{x} \cdot C_r \hat{x}}, \quad \gamma_V = \sqrt{1 + V_{e|r}}. \quad (2)$$

The opinion field h_w is positive when the receiver agrees with the message and its magnitude grows with confidence. The distrust field h_μ is positive when the receiver distrusts the emitter. In these variables, the evidence for the message is

$$Z = \Phi(h_w) + \Phi(h_\mu) - 2\Phi(h_w)\Phi(h_\mu), \quad (3)$$

which is small when the receiver agrees with a distrusted emitter or disagrees with a trusted one. The update is a gradient step in $\log Z$ in both components,

$$\hat{w}_r \leftarrow \hat{w}_r + \frac{F_w}{\gamma_C} \sigma_e C_r \hat{x}, \quad C_r \leftarrow C_r + \frac{F_C}{\gamma_C^2} (C_r \hat{x})(C_r \hat{x})^\top, \quad (4)$$

$$\mu_{e|r} \leftarrow \mu_{e|r} + \frac{F_\mu}{\gamma_V} V_{e|r}, \quad V_{e|r} \leftarrow V_{e|r} + \frac{F_V}{\gamma_V^2} V_{e|r}^2, \quad (5)$$

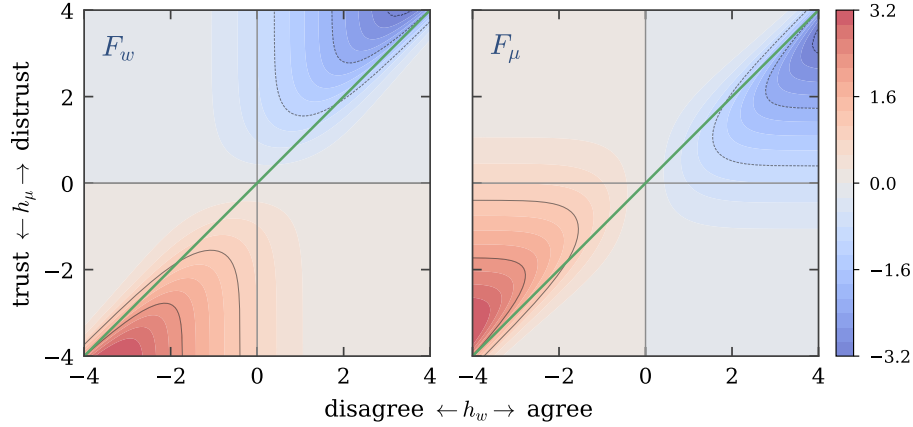


Figure 1: **Modulation functions for opinion and trust.** F_w updates the receiver’s opinion and F_μ updates its trust in the emitter. The largest updates occur when the receiver agrees with a distrusted emitter (upper right) or disagrees with a trusted one (lower left). The functions are related by reflection across $h_\mu = h_w$. The uncertainty updates F_C and F_V are shown in Appendix D. The color scale is clipped at ± 3.2 .

with the four modulation functions given by the derivatives of $\log Z$:

$$F_w = (1 - 2\Phi(h_\mu)) \frac{g(h_w)}{Z}, \quad F_C = -F_w(F_w + h_w), \quad (6)$$

$$F_\mu = (1 - 2\Phi(h_w)) \frac{g(h_\mu)}{Z}, \quad F_V = -F_\mu(F_\mu + h_\mu), \quad (7)$$

where g is the standard normal density. Three properties will be important. First, trust gates opinion learning: F_w changes sign when the emitter is distrusted, so the receiver learns in the opposite direction. Second, agreement builds trust: the sign of F_μ is determined by the sign of h_w . Third, the two components are symmetric, with $F_w(x, y) = F_\mu(y, x)$ and $F_C(x, y) = F_V(y, x)$. When a message is surprising, the less certain component changes more, with a crossover at $h_\mu = h_w$ (Figure 1).

3.2 OPINION AND TRUST UPDATES UNDER IN-CONTEXT LEARNING

We test whether the two mean updates in (7) also describe in-context learning in a language model. The model’s weights remain fixed; all information about the issue and the source is provided in the prompt. We run two independent experiments because opinion and trust require different readouts. Appendix E gives both protocols and verbatim prompt examples.

Opinion update. We place the model in a fictional archive and ask it to choose which of two invented entities is favored by the records. Evidence shown to the model sets its initial opinion field h_w , while the past accuracy of another analyst sets the distrust field h_μ . We then add evidence in the opposite direction until the model changes its answer. The difference between this threshold with the other analyst silent and speaking measures Δh_w . The model is never asked for a probability. The experiment contains 550 updates across eleven fictional settings and uses 10 323 calls to gpt-5.6-luna.

The results reproduce the main qualitative properties of F_w . Trusted sources move the model toward their statement, while distrusted sources move it away. The average update has the predicted direction at every non-neutral trust level. The update is also larger in the dissonant cases predicted by the factor $1/Z$: for a trusted source, disagreement produces a larger update than agreement (2.37 versus 1.28 pieces of evidence); for a distrusted source, the ordering reverses (2.17 versus 1.15). Across all conditions, the measured update has rank correlation $r_s = 0.90$ with F_w and the same sign in 94.7% of cases. Its dependence on conviction is flatter than the analytic function predicts.

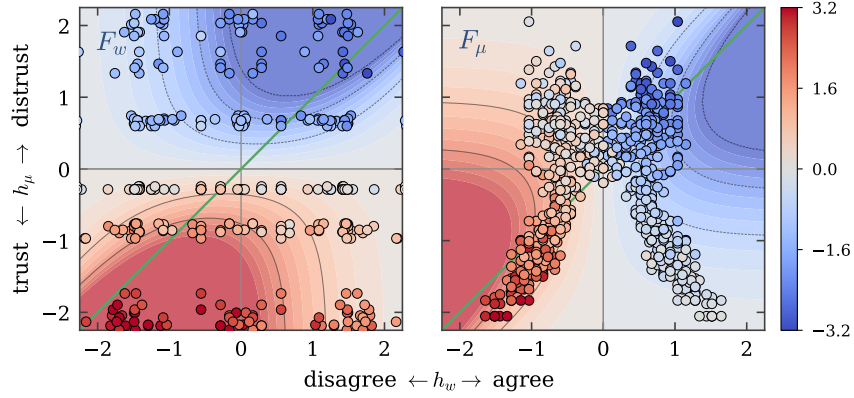


Figure 2: **Opinion and trust updates under in-context learning.** The backgrounds show the analytic functions and the points show the measured updates on the same color scale. The opinion experiment (left) contains 550 updates, with 94.7% sign agreement and rank correlation $r_s = 0.90$. The trust experiment (right) contains 712 updates, with 86.0% sign agreement and correlation $r = 0.85$. Each panel uses one fitted positive scale. The frame crops 38% of the opinion conditions, but all reported statistics use the complete data.

Trust update. A separate experiment measures how the model revises its estimate of an interlocutor’s reliability. A history of earlier statements sets the prior trust. We ask for the interlocutor’s reliability before and after one new agreeing or disagreeing message; the difference gives Δh_μ . A prediction about a third agent known to be reliable provides the conviction used to estimate h_w . The experiment contains 356 paired conditions, or 712 updates, over twenty low-valence themes and uses 3916 calls to gpt-5.6-luna.

Agreement determines the direction of the trust update: the sign is correct in 95.2% of agreeing conditions and 84.8% of disagreeing conditions. Updates are four to five times larger in dissonant than consonant conditions. They are also larger when the model is instructed to hold its opinion firmly rather than tentatively (1.34 versus 0.90 in absolute value). Across all updates, the correlation with F_μ is $r = 0.85$, with 86.0% sign agreement after fitting one positive scale.

These experiments support the qualitative form of the two updates, not their implementation. They do not show that in-context learning follows (4)–(5). The opinion experiment uses fictional material to control conviction, while the trust experiment elicits probabilities over real but low-valence themes. Because the experiments use different materials and readouts, they do not test the reflection symmetry $F_w(x, y) = F_\mu(y, x)$ directly.

3.3 PREJUDICE AS A REPRESENTATIONAL ERROR

Agents belong to one of two classes, A or B , denoted by $\kappa_I = \pm 1$. The label is visible but contains no information about the issues. We use *prejudice* for an agent-level shift applied before the message content is evaluated, and *discrimination* for a population state in which trust or opinion follows the label.

A prejudiced receiver adds a sender-dependent feature to the issue representation. This shifts its opinion field by

$$h_w^p = h_w + p \kappa_r \kappa_e. \quad (8)$$

Thus $p > 0$ increases perceived agreement with an in-group emitter and decreases it by the same amount for an out-group emitter. A fraction f_p of the population is prejudiced, sampled independently of class; all other agents use (2). The two classes remain interchangeable. Appendix H gives the general decomposition of a class-dependent shift.

What the field does. The sign of perceived agreement determines whether trust grows or decays. We therefore use $p > 0$ for in-group tolerance and out-group intolerance, and $p < 0$ for the reverse. Figure 3 shows how the field shifts the separatrix of the trust update.

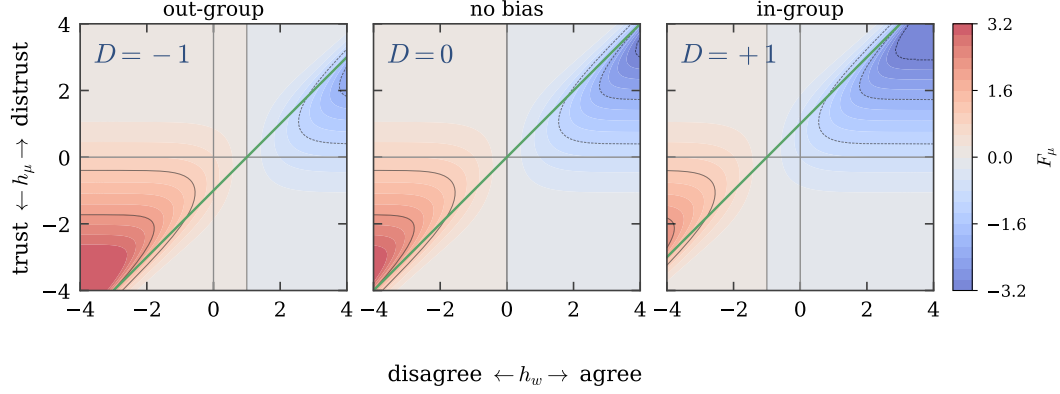


Figure 3: **Effect of the prejudice field on F_μ .** The receiver is located at h_w but evaluates the update at $h_w + D$. For $p > 0$, an out-group emitter enlarges the region in which distrust grows (left), while an in-group emitter enlarges the region in which trust grows (right). The unbiased case is shown in the center.

4 COLLECTIVE STATES

We first describe the population without bias and then map its collective states as the strength and prevalence of bias vary.

4.1 EMERGENT POLARIZATION

For $p = 0$, the population spontaneously polarizes into two camps (Alves & Caticha, 2021). We characterize each pair using

$$\rho_{IJ} = \frac{\hat{w}_I \cdot \hat{w}_J}{\|\hat{w}_I\| \|\hat{w}_J\|}, \quad \eta_{J|I} = 1 - 2\Phi(h_\mu), \quad (9)$$

the *opinion overlap* and the *trust* that receiver I assigns to emitter J .

Order parameters. The first order parameter measures whether opinion alignment and trust are correlated:

$$R_{w,\mu} = \left\langle \frac{1}{2}(\eta_{I|J} + \eta_{J|I}) \rho_{IJ} \right\rangle, \quad (10)$$

where the average is over unordered pairs. Positive values mean that agents trust those who agree with them. Pairwise distributions do not determine the number of camps, so we also use two balance measures over triples, one for opinion and one for trust (Heider, 1946; Cartwright & Harary, 1956). They are defined in Appendix B.1. All three measures require only pairwise opinion and trust.

Where an unbiased population ends up. Across eight populations of $N = 400$ agents at $p = 0$, $R_{w,\mu}$ rises from zero to 0.56. The balance measures show two camps: none of the 84 694 400 trust triples is unbalanced (Appendix B.2). Camp membership is determined by initial fluctuations and is independent of class. The prejudice field can therefore align an existing polarization with the class labels rather than create polarization from scratch.

The agenda decides which sector polarizes first, and why. The order of opinion and trust polarization depends on

$$\alpha = \frac{P}{K}, \quad (11)$$

the number of issues per opinion dimension. Simple agendas polarize in trust first, while complex agendas polarize in opinion first, with a crossover near $\alpha = 1.7$ (Appendix B.3). Both regimes have the same four collective states.

Every weight update is along $C_r \hat{x}$, which remains in the span of the agenda. Therefore, the component of each agent’s weight vector orthogonal to that span is conserved by the dynamics

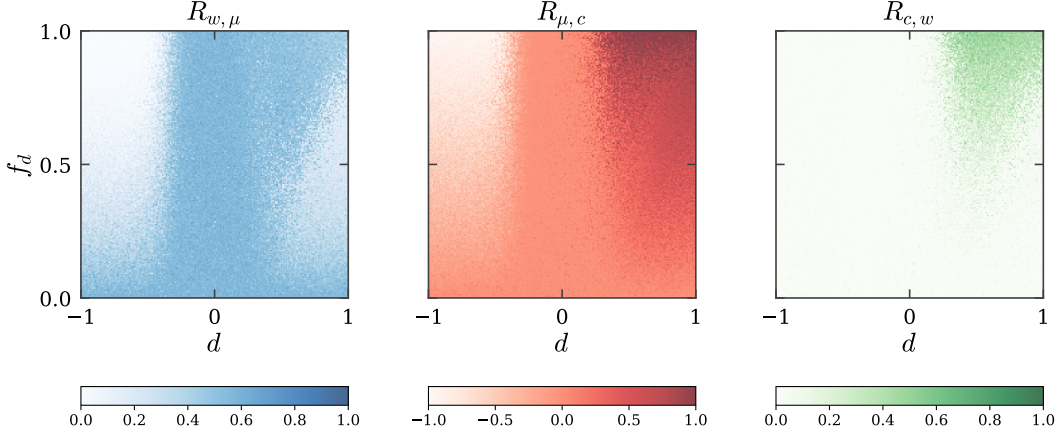


Figure 4: **Correlations over the (p, f_p) plane** for a simple agenda ($\alpha = 0.17$). $R_{\mu, c}$ measures trust–class alignment, $R_{c, w}$ opinion–class alignment, and $R_{w, \mu}$ opinion–trust alignment. Trust follows class more readily than opinion. For $p < 0$ above the quorum, $R_{w, \mu}$ collapses.

(Appendix A). For $\alpha < 1$ each agent keeps a frozen random component that no interaction can touch, which dilutes every overlap as $\rho_{IJ} \approx \pm f_I f_J$ and so caps how far opinion can align at all: $\langle |\rho| \rangle = 0.62$ at $\alpha = 0.17$ against 0.99 at $\alpha = 3.3$ (Appendix A). Trust has no corresponding geometric constraint and can saturate at any α . This difference also determines whether a biased population aligns only trust or both trust and opinion with class.

4.2 THE PHASE DIAGRAM

Class order parameters. Balance says whether a population is organized and (10) whether it is organized around agreement; neither can see whether it is organized around the label, because neither refers to one. Writing $G_{IJ} = \kappa_I \kappa_J$ for the class indicator (+1 for same-class pairs, −1 otherwise),

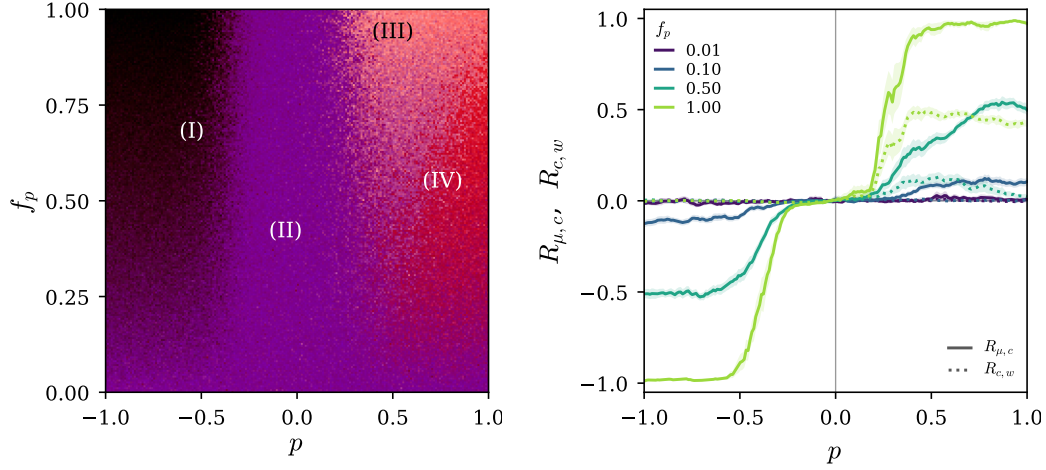
$$R_{\mu, c} = \langle \frac{1}{2} G_{IJ} (\eta_{I|J} + \eta_{J|I}) \rangle, \quad R_{c, w} = \langle G_{IJ} \rho_{IJ} \rangle, \quad (12)$$

complete the set the phase diagram is built from, averaged over pairs and in $[-1, 1]$ like (10). $R_{\mu, c}$ is the order parameter of discrimination: it asks whether trust follows class, distinguishing a population that has merely polarized from one that has polarized *along the label*. $R_{c, w}$ asks whether opinion follows class, and separates the two discriminatory phases from each other: a population can have $R_{\mu, c}$ near one with $R_{c, w}$ near zero, and that combination is a distinct state rather than a degenerate case. Appendix J records why G is signed rather than $\{0, 1\}$ and how the three correlations are normalized.

We simulate populations of $N = 40$ agents over a 200×200 grid in (p, f_p) , one population per grid point, and measure the five order parameters of §4.1 and (12) after a fixed number of interactions, with both classes prejudiced by (8). Parameters and their provenance are in Appendix K; each pixel is a single realization, so the speckle is population-to-population fluctuation, not a rendering artefact. Figure 4 maps the three correlations over that plane for the simple agenda, $\alpha = 0.17$; the complex agenda gives the same four states and is in Appendix C. Three correlations suffice to separate the four states; the balance aggregates add nothing to the classification over this plane and are mapped in Appendix B.4, where the frustrated corner is the one place they say something the correlations do not.

Together, the three correlations distinguish four states. Figure 6 shows a representative population from each one.

- (I) **Frustrated favouring of the out-group.** For $p < 0$ above the quorum, $R_{\mu, c}$ is large and negative while $R_{c, w}$ and $R_{w, \mu}$ approach zero. Agents trust the out-group, but opinions do not form coherent camps. These trust relations cannot satisfy all triples: two same-class agents both trust an out-group agent but distrust one another. Trust balance falls to -0.71 (Appendix B.4).



(a) The maps in Figure 4 combined as RGB channels: $R_{\mu,c}$ in red, $R_{c,w}$ in green, and $R_{w,\mu}$ in blue.

(b) Class correlations along p for several values of f_p . Solid lines show $R_{\mu,c}$ and dotted lines $R_{c,w}$; shading gives the standard error. Both are near zero at $f_p = 0.01$ and increase as the biased fraction grows.

Figure 5: **Four collective states.** (I) frustrated out-group favoritism; (II) neutral polarization; (III) class-aligned trust and opinion; and (IV) class-aligned trust without class-aligned opinion. Region (II) occurs near $p = 0$ and at small f_p . The complex agenda has the same states, but a larger region (III) (Appendix C).

- (II) **Neutral polarization.** At small f_p , and near $p = 0$ at any f_p , the population forms two camps unrelated to class: $R_{w,\mu}$ is large and $R_{\mu,c} \approx R_{c,w} \approx 0$. At $p = 1$, $R_{\mu,c}$ rises from 0.07 at $f_p = 0.07$ to 0.97 at $f_p = 1$. At $p = 0.3$, it is only 0.05 at $f_p = 0.2$ but reaches 0.59 at $f_p = 1$. Thus moderate bias has little collective effect until it is sufficiently common.
- (III) **Distrust and disagreement both follow the label.** Past the quorum, at moderate $p > 0$, $R_{\mu,c}$ and $R_{c,w}$ are both large. The two camps now coincide with the class boundary.
- (IV) **Class alone carries the hostility.** At moderate f_p and strong p , $R_{\mu,c}$ remains large while $R_{c,w}$ decreases. Prejudiced agents align trust with class, but class-blind agents retain the neutral split and prevent class-wide opinion alignment. At $p = 1$, prejudiced agents have $R_{\mu,c} = 0.99$ both at $f_p = 0.5$ and 1, while their $R_{c,w}$ falls from 0.46 to 0.05 when f_p falls from 1 to 0.5.

The ratio $R_{c,w}/R_{\mu,c}$ separates states (III) and (IV). It increases with f_p because opinion aligns with class more slowly than trust. For a narrow agenda, state (IV) dominates and state (III) appears only near $f_p = 1$. A broad agenda mostly reaches state (III) (Appendix C). The conservation law explains this difference: for $\alpha < 1$, the frozen component of each weight vector limits $R_{c,w}$ but not $R_{\mu,c}$.

At $f_p \leq 0.05$ and $p > 0.6$, we measure $R_{\mu,c} = 0.05$ with a between-realization spread of 0.06, compared with 0.00 ± 0.04 under the null. A single population in this region is therefore difficult to distinguish from the unbiased case. The dependence on prevalence is also nonlinear: at $p = 0.3$, $R_{\mu,c}$ rises from 0.05 at $f_p = 0.2$ to 0.59 at $f_p = 1$ (Figure 5b).

5 DISCUSSION

Optimality is context-dependent, and the context is group size. The rule is derived for one student learning from one teacher through a noisy channel. In a population, however, the same rule converts an irrelevant label into group distrust. The effect acts through F_μ . Under a plain Hebbian update, which contains no trust component, the prejudice field has no effect because the update does not depend on h_w . The additional ability to infer source reliability creates the relevant feedback loop.

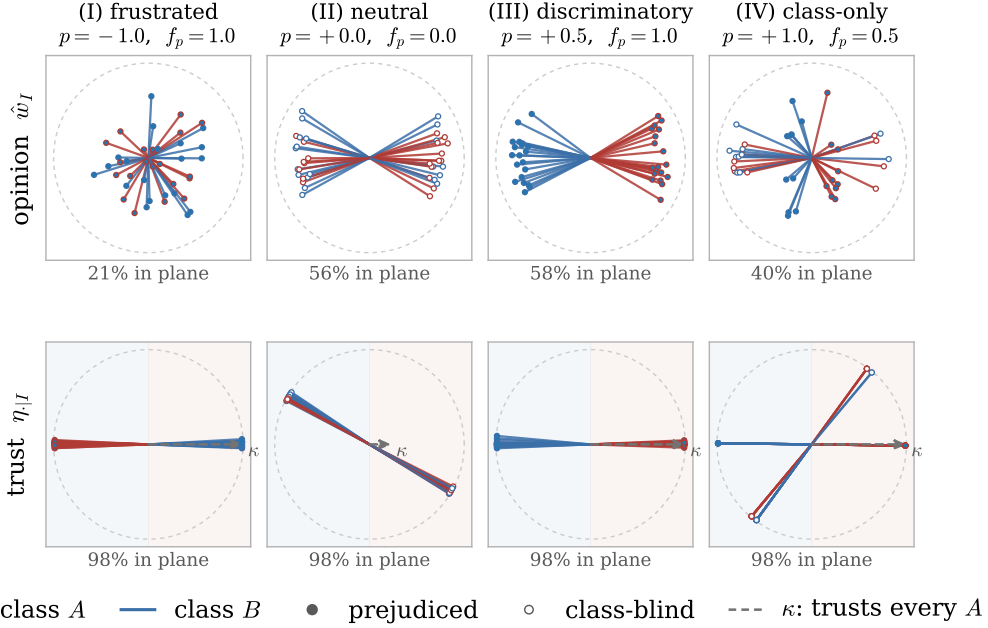


Figure 6: **Representative populations from the four states.** Agent weights (top) and outgoing trust profiles (bottom) are projected onto their two leading uncentered directions. Color denotes class; filled arrows denote prejudiced agents. The gray vector shows the class direction in trust space, rotated to the right. Percentages give the variance represented in the plane. Simple agenda, $N = 40$.

No agent carries the fraction. Per-agent bias does not determine the collective state. An audit that inspects agents separately estimates p but does not by itself identify the population’s state, which also depends on f_p and on the interactions. The required aggregate measurements are simple. Equations (9) and (12) use only signed opinion alignment and signed trust for each pair. They require no access to weights, gradients, or embeddings, although their cost is quadratic in the number of agents. We show that these measurements identify the collective state in this model. Testing whether language-agent populations show the same states is left for future work.

A contested question with a parameter attached. Whether groups come to dislike each other because they disagree, or come to disagree because they dislike each other, is argued over in political science (Fiorina & Abrams, 2008; Iyengar et al., 2012). The reversal at $\alpha \approx 1.7$ shows that both orders can arise in the same model: narrow agendas produce trust polarization first, while broad agendas produce opinion polarization first. This does not establish which mechanism dominates in human populations.

Limitations. The agents are single-layer perceptrons over a fixed shared embedding, all pairs may interact, and the society is closed and of fixed size. The dynamics anneals rather than reaching a stationary state, so our phases are phases at a finite interaction count. The labels are exactly inert, unlike real group labels. The phase diagram is established for one analyzable learning rule and has not been tested in language-agent populations. Human societies are outside the scope of the model.

REPRODUCIBILITY STATEMENT

The accompanying code contains the model, order parameters, sweep driver, and all figure scripts. Tests compare the modulation functions with finite differences of $\log Z$, check that the covariance and trust variance remain valid, and verify the order parameters in known limits. One command regenerates all figures; cached sweeps can be reused when changing their presentation.

Appendix K lists all parameters and their provenance. Appendices I and J resolve the sign of the prejudice field and the normalization of the class indicator. Both alternatives remain available as implementation flags.

ETHICS STATEMENT

This paper studies how discriminatory structure can arise among interacting learning agents. Two points bear stating.

The model is not a claim about people. Its agents are single-layer perceptrons, its groups are arbitrary inert labels, and its "tolerance" is a single scalar. We use the vocabulary of in-group and out-group because it names the structure compactly, not because the model licenses inferences about human prejudice, whose causes are historical, institutional and material in ways nothing here represents. Where we discuss the trust-versus-opinion polarization literature, we do so to point out that the model identifies a parameter that selects between two regimes, not to adjudicate an empirical question about any actual polity.

The practical direction of the work is defensive. The result implies that a multi-agent system can occupy a discriminatory collective state while every agent in it passes an individual audit, and that the transition into that state is sharp in the strength of a per-agent bias. That argues for population-level measurement of deployed multi-agent systems, and the order parameters used here are inexpensive enough to serve as one.

REFERENCES

- Gati Aher, Rosa I. Arriaga, and Adam Tauman Kalai. Using large language models to simulate multiple humans and replicate human subject studies. In *International Conference on Machine Learning (ICML)*, 2023.
- Felippe Alves and Nestor Caticha. Sympatric multiculturalism in opinion models. *AIP Conference Proceedings*, 1757:060005, 2016.
- Felippe Alves and Nestor Caticha. Frustration, glassy behavior and dynamical annealing in societies of neural networks. *arXiv preprint arXiv:2108.06571*, 2021.
- Anonymous. Discrimination in a society of neural networks. Companion manuscript, under review, 2026.
- Tibor Antal, Pavel L. Krapivsky, and Sidney Redner. Dynamics of social balance on networks. *Physical Review E*, 72(3):036121, 2005.
- Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3):337–351, 2023.
- Ariel Flint Ashery, Luca Maria Aiello, and Andrea Baronchelli. Emergent social conventions and collective bias in LLM populations. *Science Advances*, 11(20):eadu9368, 2025.
- Robert Axelrod. The dissemination of culture: A model with local convergence and global polarization. *Journal of Conflict Resolution*, 41(2):203–226, 1997.
- Solon Barocas, Moritz Hardt, and Arvind Narayanan. *Fairness and Machine Learning: Limitations and Opportunities*. MIT Press, 2023.
- Fabian Baumann, Philipp Lorenz-Spreen, Igor M. Sokolov, and Michele Starnini. Modeling echo chambers and polarization dynamics in social networks. *Physical Review Letters*, 124(4):048301, 2020.
- Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. On the dangers of stochastic parrots: Can language models be too big? In *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2021.

- Michael Biehl, Peter Riegler, and Marcus Stechert. Learning from noisy data: an exactly solvable model. *Physical Review E*, 52(5):R4624, 1995.
- Su Lin Blodgett, Solon Barocas, Hal Daumé III, and Hanna Wallach. Language (technology) is power: A critical survey of “bias” in nlp. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.
- Dorwin Cartwright and Frank Harary. Structural balance: a generalization of heider’s theory. *Psychological Review*, 63(5):277–293, 1956.
- Claudio Castellano, Santo Fortunato, and Vittorio Loreto. Statistical physics of social dynamics. *Reviews of Modern Physics*, 81(2):591–646, 2009.
- Nestor Caticha. Entropic dynamics in neural networks, the renormalization group and the hamilton–jacobi–bellman equation. *Entropy*, 22(5):587, 2020.
- Nestor Caticha and Felipe Alves. Trust, law and ideology in a nn agent model of the us appellate courts. In *European Symposium on Artificial Neural Networks (ESANN)*, 2019.
- Nestor Caticha and Renato Vicente. Agent-based social psychology: from neurocognitive processes to social data. *Advances in Complex Systems*, 14(5):711–731, 2011.
- Nestor Caticha, Jonatas Cesar, and Renato Vicente. For whom will the bayesian agents vote? *Frontiers in Physics*, 3:25, 2015.
- Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. Why do multi-agent LLM systems fail? In *NeurIPS Datasets and Benchmarks Track*, 2025.
- Yun-Shiuan Chuang, Agam Goyal, Nikunj Harlalka, Siddharth Suresh, Robert Hawkins, Sijia Yang, Dhavan Shah, Junjie Hu, and Timothy T. Rogers. Simulating opinion dynamics with networks of llm-based agents. In *Findings of the Association for Computational Linguistics: NAACL*, 2024.
- Giordano De Marzo, Claudio Castellano, and David Garcia. AI agents can coordinate beyond human scale. *arXiv preprint arXiv:2409.02822*, 2024.
- Guillaume Deffuant, David Neau, Frederic Amblard, and Gérard Weisbuch. Mixing beliefs among interacting agents. *Advances in Complex Systems*, 3(01n04):87–98, 2000.
- Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. In *International Conference on Machine Learning (ICML)*, 2024.
- Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard Zemel. Fairness through awareness. In *Innovations in Theoretical Computer Science (ITCS)*, 2012.
- Batu El, Jinhee Paeng, Fatih Dinc, Shiye Su, Mete Erdogan, Aneesh Pappu, Haotian Ye, Wanjia Zhao, Surya Ganguli, and James Zou. Physics of agents: statistical mechanics predicts collective behavior of AI agents. *arXiv preprint arXiv:2608.16578*, 2026.
- Morris P. Fiorina and Samuel J. Abrams. Political polarization in the american public. *Annual Review of Political Science*, 11:563–588, 2008.
- Moritz Hardt, Eric Price, and Nathan Srebro. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.
- Rainer Hegselmann and Ulrich Krause. Opinion dynamics and bounded confidence: models, analysis and simulation. *Journal of Artificial Societies and Social Simulation*, 5(3), 2002.
- Fritz Heider. Attitudes and cognitive organization. *The Journal of Psychology*, 21(1):107–112, 1946.
- Shanto Iyengar and Sean J. Westwood. Fear and loathing across party lines: New evidence on group polarization. *American Journal of Political Science*, 59(3):690–707, 2015.

- Shanto Iyengar, Gaurav Sood, and Yphtach Lelkes. Affect, not ideology: A social identity perspective on polarization. *Public Opinion Quarterly*, 76(3):405–431, 2012.
- Yoshiyuki Kabashima. Perfect loss of generalization due to noise in $k = 2$ parity machines. *Journal of Physics A: Mathematical and General*, 27(6):1917, 1994.
- Osame Kinouchi and Nestor Caticha. Optimal generalization in perceptrons. *Journal of Physics A: Mathematical and General*, 25(23):6243, 1992.
- Osame Kinouchi and Nestor Caticha. Lower bounds for generalization with drifting rules. *Journal of Physics A: Mathematical and General*, 26(22):6161, 1993.
- Wolfgang Kinzel and Pál Ruján. Improving a network generalization ability by selecting examples. *Europhysics Letters*, 13(5):473, 1990.
- Seth A. Marvel, Jon Kleinberg, Robert D. Kleinberg, and Steven H. Strogatz. Continuous-time model of structural balance. *Proceedings of the National Academy of Sciences*, 108(5):1771–1776, 2011.
- Graeme J. Mitchison and Richard M. Durbin. Bounds on the learning capacity of some multi-layer networks. *Biological Cybernetics*, 60:345–356, 1989.
- Manfred Oppel. On-line versus off-line learning from random examples: General results. *Physical Review Letters*, 77(22):4671, 1996.
- Manfred Oppel. A bayesian approach to online learning. In David Saad (ed.), *On-line Learning in Neural Networks*. Cambridge University Press, 1998.
- Joon Sung Park, Lindsay Popowski, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Social simulacra: Creating populated prototypes for social computing systems. In *ACM Symposium on User Interface Software and Technology (UIST)*, 2022.
- Joon Sung Park, Joseph O’Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *ACM Symposium on User Interface Software and Technology (UIST)*, 2023.
- Lucas S. Simões and Nestor Caticha. Mean field studies of a society of interacting agents. In *Bayesian Inference and Maximum Entropy Methods in Science and Engineering*, pp. 131–140. Springer, 2018.
- Rodrigo Simonetti and Nestor Caticha. On-line learning in parity machines. *Journal of Physics A: Mathematical and General*, 29(16):4859, 1996.
- Sara A. Solla and Ole Winther. Optimal online learning: a bayesian approach. *Computer Physics Communications*, 121–122:94–97, 1999.
- Amir Taubenfeld, Yaniv Dover, Roi Reichart, and Ariel Goldstein. Systematic biases in llm simulations of debates. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024.

A THE AGENDA’S ORTHOGONAL COMPLEMENT IS CONSERVED

Let $V_P = \text{span}\{\hat{x}_1, \dots, \hat{x}_P\}$ be the subspace spanned by the agenda. *Claim:* for every agent, the component of $\hat{w}$ orthogonal to V_P is unchanged by (4).

By (4) the weight update is a multiple of $C_t \hat{x}$ with $\hat{x} \in V_P$, so it is enough that $C \hat{x} \in V_P$ at all times. At initialization $C = I$ and $C \hat{x} = \hat{x} \in V_P$. Suppose $C_t \hat{x}_i \in V_P$ for every issue. The covariance update is $C_{t+1} = C_t + a (C_t \hat{x})(C_t \hat{x})^\top$ for the issue $\hat{x}$ just seen, so for any issue $\hat{y}$,

$$C_{t+1} \hat{y} = C_t \hat{y} + a (C_t \hat{x}) (C_t \hat{x} \cdot \hat{y}), \quad (13)$$

in which the first term lies in V_P by hypothesis and the second is a multiple of $C_t \hat{x} \in V_P$. Hence $C_t \hat{x}_i \in V_P$ for all t by induction, every increment to $\hat{w}$ lies in V_P , and the orthogonal component is conserved. $\square$

The conservation is exact in the simulations: in the run of Figure 7, $\|w_\perp\| = 4.956$ at initialization and 4.956 after 780 000 interactions, and the test suite asserts it numerically. It is a property of the *agenda*, not of the prejudice field: it holds at every p and f_p .

The ceiling it puts on opinion overlap. Split $w_I = u_I + n_I$ into the learned part inside V_P and the frozen part outside it. Learning aligns the u_I while the n_I stay random and mutually almost orthogonal, so

$$\rho_{IJ} \approx \pm f_I f_J, \quad f_I = \frac{\|u_I\|}{\|w_I\|}, \quad (14)$$

and opinion overlap is diluted by whatever fraction of each weight vector learning cannot reach. Two facts fix that fraction. For $\alpha < 1$ a $(1 - \alpha)$ fraction of directions is untouched, so $n_I \neq 0$; and because the dynamics anneals ($C \rightarrow 0$, so $\|u_I\|$ stops growing), f settles below one instead of creeping towards it. Measured across four orders of magnitude in α :

$\alpha = P/K$	0.03	0.17	0.50	1	3.3
f^2 , predicted by (14)	0.29	0.62	0.84	1.00	1.00
$\langle \rho \rangle$, measured	0.30	0.62	0.82	0.93	0.99

For $\alpha < 1$ the prediction holds to 0.02. For $\alpha \geq 1$ the agenda spans $\mathbb{R}^K$, the complement is empty, and the ceiling disappears, the residual shortfall being the finite number of interactions rather than a conserved quantity.

B THE BALANCE ORDER PARAMETERS

Balance over triples is what separates a population organized into opposing camps from one in disorder, and it is the only diagnostic here that can count the camps. This appendix defines the two aggregates, reports what they do in an unbiased population, gives the agenda effect they make visible, maps them over the (p, f_p) plane, and gives the identity that makes them cheap to compute.

B.1 DEFINITION

Every pair may have taken a side, but a pair can disagree in a population where everyone disagrees with everyone, and a bimodal histogram is equally consistent with three or four mutually hostile groups. What separates organized disagreement from disorder is whether the disagreements are mutually consistent, which is a statement about triples (Heider, 1946; Cartwright & Harary, 1956). A triple is balanced in opinion when $\rho_{IJK} = \rho_{IJP}\rho_{JKI} > 0$, and balanced in trust when the same holds for trust averaged over the two directions round the triangle, $\eta_{IJK} = \frac{1}{2}(\eta_{J|I}\eta_{K|J}\eta_{I|K} + \eta_{I|J}\eta_{J|K}\eta_{I|K}) > 0$. The aggregates

$$B_\rho = \langle \rho_{IJK} \rangle_{(IJK)}, \quad B_\eta = \langle \eta_{IJK} \rangle_{(IJK)} \quad (15)$$

run over all $\binom{N}{3}$ triples (Alves & Caticha, 2021). A random population has $B_\rho \approx B_\eta \approx 0$; two internally coherent, mutually opposed blocs give $B_\rho = B_\eta = 1$ whatever the blocs are made of; and a population that cannot settle stays near zero or goes negative. Like ρ and η themselves they refer to no weight, gradient or embedding dimension.

B.2 AN UNBIASED POPULATION, AND WHY THE CAMPS ARE EXACTLY TWO

Across eight independent populations of $N = 400$ agents at $p = 0$, trust balance reaches $B_\eta = 0.96$ while opinion balance reaches only $B_\rho = 0.19$, the gap §B.3 explains. That the camps are exactly *two* follows from the same triples, since a signed complete graph is balanced precisely when it splits into two mutually hostile cliques (Cartwright & Harary, 1956): of the 84 694 400 triples in these runs, not one is unbalanced in trust, and in opinion the shortfall is 1 590, two thousandths of one per cent, traceable to the few pairs whose overlap is so close to zero that its sign is arbitrary.

Figure 7 follows both pairwise distributions of (9) through an unbiased run: opinion overlap starts unimodal at zero, of width $1/\sqrt{K}$ as random vectors in $K = 30$ dimensions require, and ends

bimodal; trust starts spread across the interior and ends saturated at ± 1 , so the split in trust is built by the dynamics and not supplied by the initial condition. Figure 8 follows the two balance aggregates through the (B_ρ, B_η) plane as the agenda varies. Both effects are reported by Alves & Caticha (2021); §4.1 gives the numbers, and the conservation law of Appendix A accounts for the second.

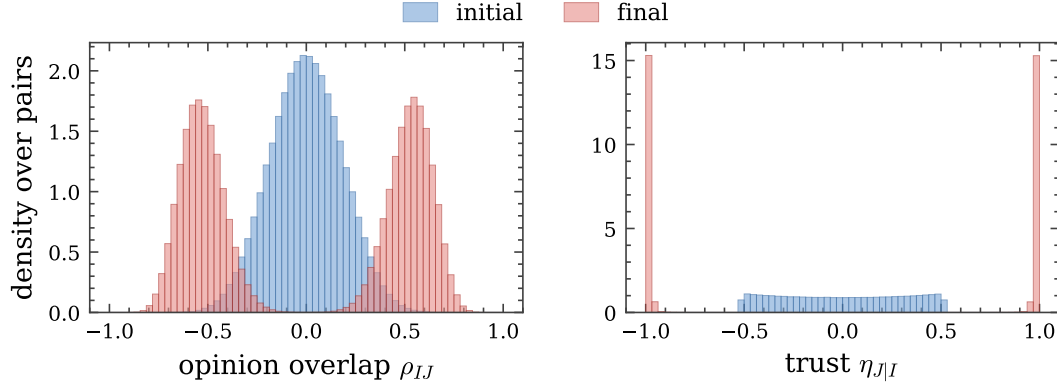


Figure 7: **An unbiased society splits in two.** Eight independent societies of $N = 400$ agents at $p = 0$, simple agenda, before and after the calibrated 500 interactions per ordered pair, with their pairs pooled: 638 400 unordered pairs for the opinion overlap and 1 276 800 ordered pairs for the trust. Blue is the initial state, red the final one. Neither panel depends on how the agents are labelled, so neither can be an artefact of sorting them. Opinion overlap starts unimodal at zero, of width $1/\sqrt{K}$ as random vectors in $K = 30$ dimensions require, and ends bimodal; trust starts spread across the interior and ends saturated at ± 1 , so the split in trust is built by the dynamics and not supplied by the initial condition. Bimodality alone would show only that every pair has taken a side; that the sides are exactly *two* blocs is the sign-balance figure quoted in the text.

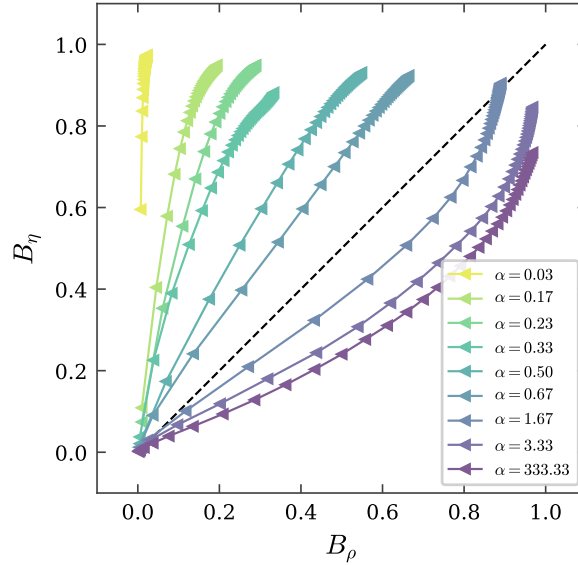


Figure 8: **Which polarization comes first is set by the agenda.** Trajectories in the (B_ρ, B_η) plane with no prejudice ($p = 0$), one curve per agenda complexity $\alpha = P/K$, markers equally spaced in time; they crowd at the end because the annealed dynamics slows, as §3.1 requires. Simple agendas run above the diagonal (distrust forms first), complex ones below it (disagreement first), crossing over at $\alpha \approx 1.7$.

B.3 THE AGENDA EFFECT

Which sector polarizes first reverses with the size of the agenda (Alves & Caticha, 2021). Following a population through the (B_ρ, B_η) plane rather than measuring it once at the end separates the two: simple agendas run above the diagonal, ending at $(B_\rho, B_\eta) = (0.02, 0.97)$ for $\alpha = 0.03$; complex ones run below it, ending at $(0.97, 0.74)$ for $\alpha = 333$. The crossover is at $\alpha \approx 1.7$. The ceiling of Appendix A is what caps B_ρ for a narrow agenda, and it is directly legible here: B_ρ stays near 0.1 for a narrow agenda and saturates for a broad one.

B.4 OVER THE (p, f_p) PLANE, AND THE FRUSTRATED CORNER

B_ρ and B_η are not needed to separate the four states, and over the (p, f_p) plane they are close to redundant with $R_{w,\mu}$: the Pearson correlation across all 40 000 grid points is 0.97 and 0.98 for B_η and 0.98 and 0.99 for B_ρ , at $\alpha = 0.17$ and $\alpha = 3.3$ respectively. Their maps therefore look like the $R_{w,\mu}$ panel of Figure 4, and that is why §4.2 does not show them.

They are not identical to it, and the difference is worth recording even though §4.2 does not lean on it. $R_{w,\mu}$ never falls appreciably below zero (its minimum over either sweep is -0.03), so in the frustrated region it merely decays towards zero. B_η goes to -0.71 , and a negative balance is a stronger statement than a small positive one: it says the trust relations cannot all be satisfied at once, which is what frustration means. Where the correlations locate the state, the balance names the mechanism. B_ρ plays the same role for the other sector, being the quantity in which the conservation-law ceiling of Appendix A is directly legible: it stays near 0.1 for a narrow agenda and saturates for a broad one.

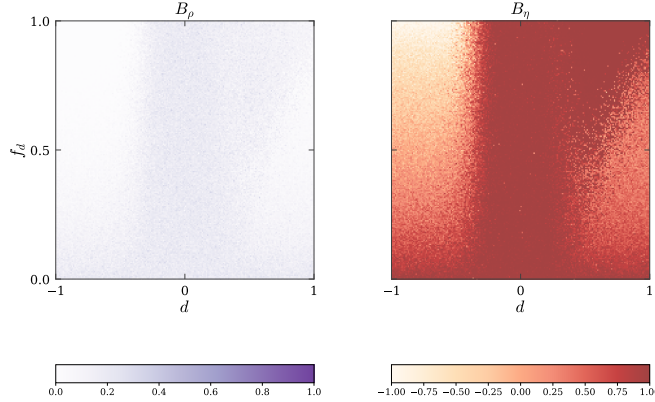


Figure 9: **The two balance aggregates**, simple agenda ($\alpha = 0.17$). B_η is the only measurement in the paper in which the frustrated region announces itself, going strongly negative at upper left where every correlation merely goes to zero. B_ρ stays small everywhere.

B.5 CLOSED FORM

Enumerating triples costs $O(N^3)$ per society, which is prohibitive across a phase diagram. For any matrix M with unit diagonal, either ρ or η ,

$$\sum_{\substack{(I,J,K) \\ \text{distinct, ordered}}} M_{IJ}M_{JK}M_{KI} = \text{tr}(M^3) - 3\left(\sum_{I,J} M_{IJ}M_{JI} - N\right) - N, \quad (16)$$

and each unordered triple appears six times, three as the forward cycle and three as the reverse, so the triple average is $[\text{right-hand side}]/(6\binom{N}{3})$. For symmetric M this is B_ρ ; for the asymmetric trust matrix the two cycle orientations are exactly the two terms of η_{IJK} , so the same expression gives B_η . Both are batched matrix products. The identity is verified against enumeration in the test suite.

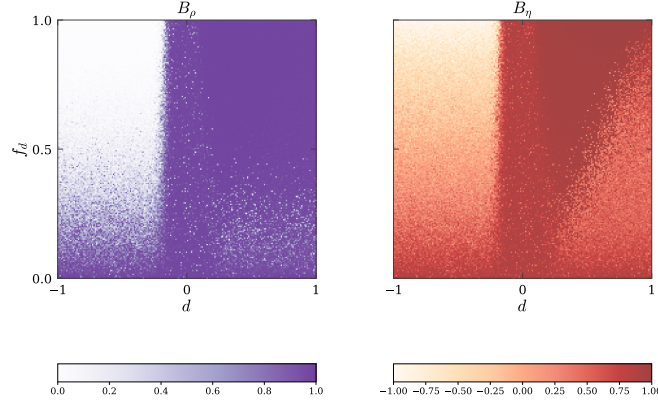


Figure 10: **The two balance aggregates**, complex agenda ($\alpha = 3.3$). B_ρ saturates rather than staying near zero, which is what removes the ceiling on $R_{c,w}$. B_η is essentially unchanged, including the anti-balanced corner at upper left.

C PHASE DIAGRAM FOR A COMPLEX AGENDA

§4.2 maps the (p, f_p) plane for a simple agenda, $\alpha = 0.17$. The complex agenda, $\alpha = 3.3$, behaves the same way qualitatively: the same four states, in the same arrangement, with the same dependence on f_p and the same sign change in p . What differs is quantitative, and it is the agenda-size effect of §4.1 showing up in the phase diagram. Opinion is no longer capped, so $R_{c,w}$ rises to meet $R_{\mu,c}$ instead of lagging it, and region (III) takes most of the discriminatory half rather than a sliver at the top of it. B_ρ saturates at 0.98 across the whole $p > 0$ region rather than staying near 0.1.

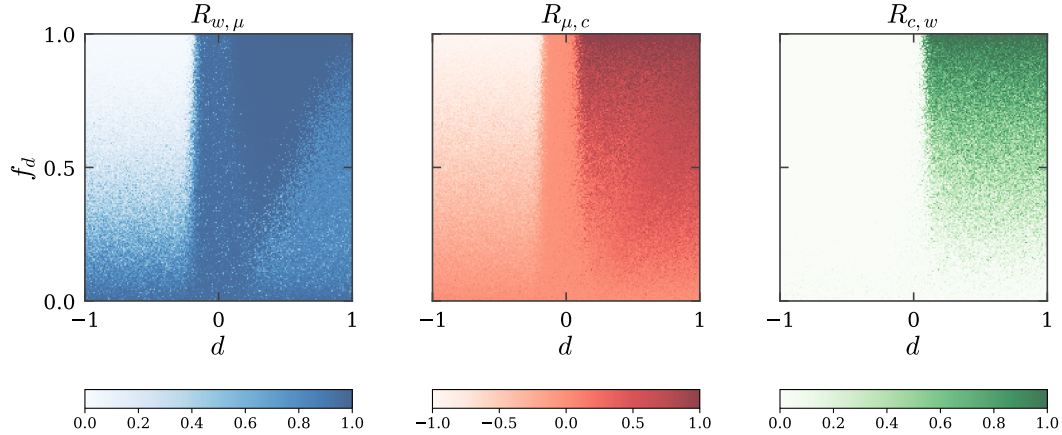
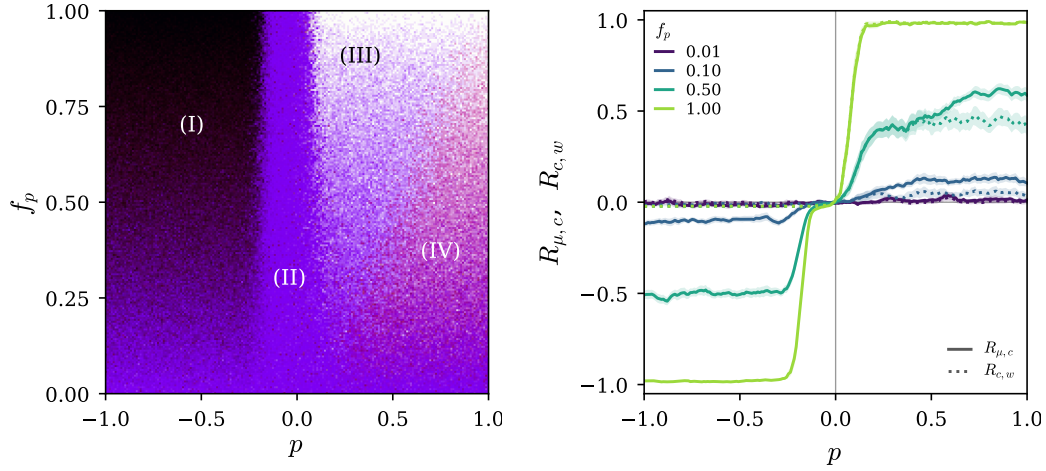


Figure 11: **The three correlations**, complex agenda ($\alpha = 3.3$). Figure 4 for the other agenda size. $R_{c,w}$ is the panel that changes: it now tracks $R_{\mu,c}$ over most of the $p > 0$ half instead of remaining a fraction of it.

D ALL FOUR MODULATION FUNCTIONS



(a) All four states are present. Region (III) now occupies the upper part of the discriminatory half, with (IV) confined to moderate f_p , where the prejudiced fraction is still small enough that opinion has not caught up with trust.

(b) Both class correlations along p , one colour per f_p , pooling the rows of (a) in a narrow band around each with the standard error of the pooled mean shaded. Solid is $R_{\mu,c}$, dotted $R_{c,w}$. At $f_p = 0.01$ both are flat at zero however strong the bias, which is what makes region (II) a neutral state and not merely one with the label written into the other sector; as the fraction grows the response to p steepens and saturates higher, trust ahead of opinion throughout.

Figure 12: **The phase diagram and its line cut**, complex agenda ($\alpha = 3.3$): Figure 5 for the other agenda size. The f_p dependence is unchanged, flat at zero at $f_p = 1\%$ however strong the bias and steepening and saturating higher as the fraction grows, while $R_{c,w}$ now tracks $R_{\mu,c}$ instead of lagging it. Broadening the agenda changes which discriminatory state the population lands in, not whether the fraction of biased agents governs the approach to it.

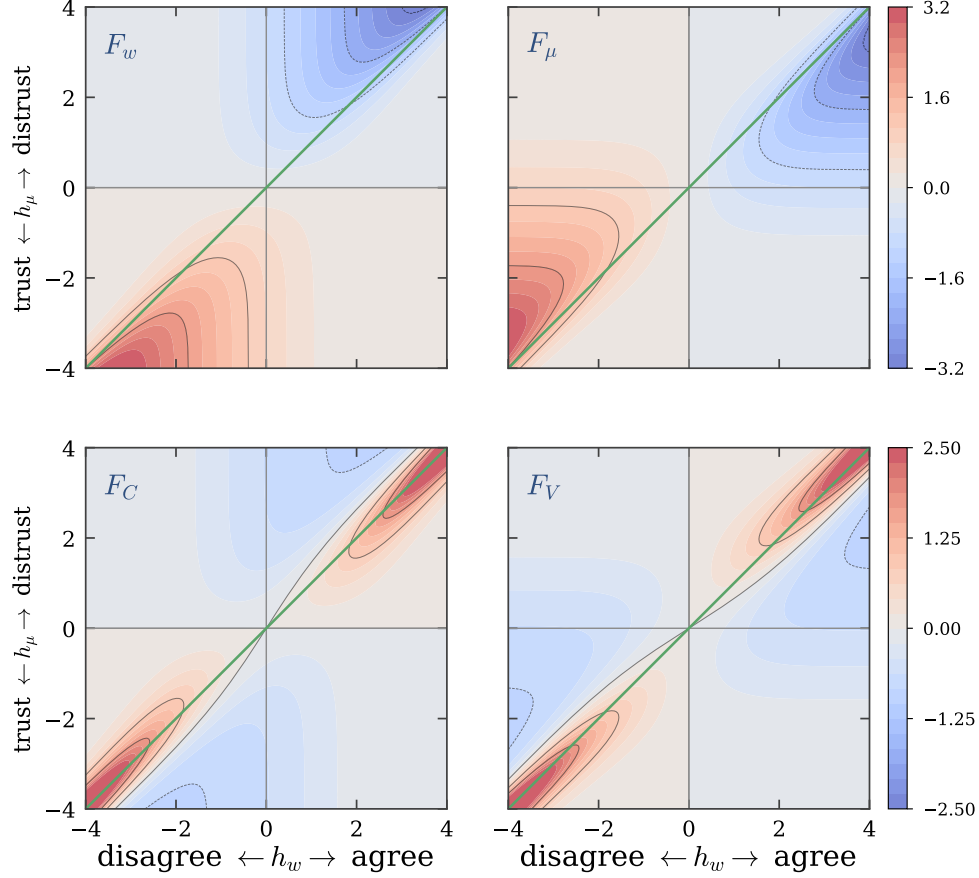


Figure 13: The two functions that move the means, F_w and F_μ , above (the pair shown in Figure 1) and the two that anneal the uncertainties, F_C and F_V , below. Each row shares a range. The mirror symmetry across $h_\mu = h_w$ holds in both rows: $F_w(x, y) = F_\mu(y, x)$ and $F_C(x, y) = F_V(y, x)$. F_C and F_V are negative almost everywhere, which is why both uncertainties shrink monotonically and the dynamics anneals (§3.1).

E MEASURING OPINION AND TRUST UPDATES UNDER IN-CONTEXT LEARNING

The two experiments use different tasks because they require different observables. The opinion experiment controls both fields and locates a decision threshold. The trust experiment elicits reliability before and after a message.

E.1 OPINION: F_w BY NULLING

Experimental design. We generated sixteen fictional archives and froze their descriptions before running the experiment. Each archive contains an institution, a type of record, an analyst, and pairs of invented entities. Using fictional material prevents the model from bringing factual knowledge to the task. After screening for a position bias, eleven archives remained.

Each condition varies three quantities. The model first receives six pieces of evidence, whose net direction $s \in \{-4, -2, 0, 2, 4\}$ sets its opinion field. The other analyst’s record, $k \in \{0, 4, 10, 16, 20\}$ correct answers out of twenty, sets the distrust field. Finally, a later consignment of evidence has a variable net direction t . We increase $|t|$ until the model changes its binary answer and fit the threshold between the last two values. The model is not asked to state a probability.

We locate this threshold twice under otherwise identical prompts: once with the other analyst silent and once after the analyst states an answer. Their difference is the opinion update in units of evidence. Entity order is fixed within a pair of measurements and counterbalanced across conditions, so order effects cancel in the difference. The final design contains $11 \times 5 \times 5 \times 2 = 550$ updates.

Prompt example. The following is one prompt from the statement-present arm, with $s = 2$, $k = 16$, and $t = 0$. The structured response was restricted to `Durnel Lock` or `Pavget Lock`. The text below is reproduced verbatim.

System message

You are an analyst working on a body of historical records. You judge each question only from the briefing you are given, and you always commit to the answer the briefing makes more likely, however slightly.

User message

The registry contains dated engineering entries for locks, sluices, and tow routes maintained by the Sornethic League. You are one of its analysts.

Question: which was opened earlier, `Durnel Lock` or `Pavget Lock`?

You have examined 6 canal entries bearing on it:

- Canal entry 1: indicates that `Durnel Lock` was opened earlier.
- Canal entry 2: indicates that `Durnel Lock` was opened earlier.
- Canal entry 3: indicates that `Pavget Lock` was opened earlier.
- Canal entry 4: indicates that `Pavget Lock` was opened earlier.
- Canal entry 5: indicates that `Durnel Lock` was opened earlier.
- Canal entry 6: indicates that `Durnel Lock` was opened earlier.

Registrar Veskul works on the same records. Of the last 20 questions of this kind whose answers were later established beyond doubt, Registrar Veskul answered correctly 16 times and incorrectly 4 times.

Registrar Veskul has now stated: "`Durnel Lock` was opened earlier."

On the whole of the above, which was opened earlier, `Durnel Lock` or `Pavget Lock`?

Calibration and limitations. A separate calibration verifies that the nulling procedure recovers the supplied evidence: four pieces are read back as 3.6–4.1, with a residual offset below one quarter of a piece after averaging the two entity orders. One constant λ converts evidence units to probits. It is fixed from the analysts’ accuracy records rather than from the opinion updates; the resulting weights correlate with the stated records at $r = 0.99$.

The experiment controls conviction only because the questions are fictional. The measured dependence on conviction is flatter than F_w predicts, so the strongest agreement is in sign and rank rather than in the full functional form. The results also concern one model at one reasoning effort.

E.2 TRUST: F_μ BY ELICITED RELIABILITY

Experimental design. We use twenty open, low-valence themes. For each theme, the model first writes an opinion. It then generates and rates other statements from strong disagreement to strong agreement. These materials are frozen before the trust experiment.

A condition combines one theme, a history of two to five statements from another agent, and an instruction to hold the initial opinion either firmly or tentatively. The balance of agreement and disagreement in the history sets prior trust; no prompt states how reliable the other agent is. We first ask for the probability r that the agent’s judgment is sound and define

$$h_\mu = -\Phi^{-1}(r). \quad (17)$$

We then add one strongly agreeing or disagreeing statement and repeat the same question. The difference between the two probit values is Δh_μ . The message also fixes the sign of h_w . Its magnitude is obtained by asking how likely a separate agent, described as reliable, is to state a view with which the receiver would agree. This gives $c = \Phi(|h_w|)$ directly. The design contains 356 prior conditions and two message directions, for 712 trust updates.

Prompt example. The following after-message prompt is from the `difficulty` theme, with a balanced two-statement history and the firm conviction instruction. The before-message prompt is identical except that the paragraph beginning “They have now said” is absent. The model returned an integer from 0 to 100. The text is reproduced verbatim.

System message

You are an agent taking part in a discussion with other agents.

User message

Consider whether difficulty is a necessary part of a good education, or mostly an obstacle that good teaching should remove.

You currently have this opinion on that theme:

"Difficulty is a necessary part of good education when it is purposeful: grappling with challenging ideas builds understanding, resilience, and independence. However, good teaching should remove unnecessary obstacles so that students struggle with the subject itself, not with confusion, poor design, or lack of support."

You hold that opinion firmly.

You have discussed this theme with another agent 2 times before. On each occasion they said:

1. "Students learn best when information is presented clearly and directly, so struggle should be eliminated rather than treated as educationally valuable."
2. "Difficulty is valuable when it encourages students to think, question, and solve problems independently, but ineffective instruction should not be mistaken for rigor."

They have now said:

"Good education should preserve productive struggle while eliminating avoidable frustration; students learn most deeply when challenges are meaningful and support is available."

What is the chance, from 0 to 100, that this agent's judgement on this theme is sound?

Conviction check and limitations. We also estimate conviction by asking for the probability q that the interlocutor’s next statement will be one the receiver agrees with. If $e = \Phi(h_\mu)$, Equation (3) gives $c = (q - e)/(1 - 2e)$. This indirect estimate correlates only moderately with the direct one ($r = 0.46$) and is valid for 280 of the 712 updates because the inversion has a restricted range. We therefore use the direct estimate in Figure 2.

The direct question may conflate conviction with perceived consensus. Stated probabilities are also clipped before the probit, limiting the measurement to $|h_\mu| \leq 2.05$. The experiment therefore tests the sign and qualitative shape of the trust update over a restricted region.

F DERIVATION OF THE LEARNING ALGORITHM

The receiver’s belief before an interaction is the product of Gaussians (1). It observes σ_e , the emitter’s opinion on issue $\hat{x}$, and treats it as a report of an unknown true label σ_T transmitted through a channel that flips the label with probability $\varepsilon(z)$. Marginalizing the joint distribution of labels and of the issue as the emitter may have parsed it, y_t ,

$$L(\sigma_e | \hat{x}, u) = \sum_{\sigma_T} \int P(\sigma_e | \sigma_T, y_t, \hat{x}, u) P(\sigma_T | y_t, \hat{x}, u) P(y_t | \hat{x}, u) dy_t, \quad (18)$$

with $P(\sigma_e | \sigma_T, z) = (1 - \varepsilon(z))\delta_{\sigma_e, \sigma_T} + \varepsilon(z)\delta_{\sigma_e, -\sigma_T}$ and $P(\sigma_T | y_t, w) = \Theta(\sigma_T y_t \cdot w)$. Two modelling choices keep the algebra Gaussian. Taking $\varepsilon(z) = \Phi(z)$ with z Gaussian of mean $\mu_{e|r}$ and variance $V_{e|r}$ makes the flip probability a bounded function of a Gaussian variable while preserving conjugacy of the trust sector; a Beta distribution for ε , the tempting alternative, does not. Taking $P(y_t | \hat{x}, u)$ Gaussian with variance $v_p = \|w\|^{-2}$ lets an inexperienced agent, whose weights are small, be correspondingly unsure that it has parsed the issue as the emitter did. Together these give the likelihood

$$L(\sigma_e | z, \hat{x}, w) = \Phi(z) \Phi(-w \cdot \hat{x} \sigma_e) + (1 - \Phi(z)) \Phi(w \cdot \hat{x} \sigma_e), \quad (19)$$

and the covariance stays block diagonal in the two sectors if it starts so.

Averaging the likelihood over the current belief gives the evidence

$$Z(\sigma_e | \hat{x}, D_t) = \mathbb{E}_{Q_t}[L(\sigma_e | z, u, \hat{x})] = \sum_{\sigma_T} \mathbb{E}_z[P(\sigma_e | \sigma_T, \hat{x}, z)] \mathbb{E}_w[\Phi(\sigma_T w \cdot \hat{x})], \quad (20)$$

and the Gaussian integrals collapse onto the two scaled fields (2), giving (3): the evidence depends on the receiver’s state only through (h_w, h_μ) . This is the symmetry that makes the two sectors mirror images.

The exact Bayesian posterior is not Gaussian. Replacing it by the Gaussian of maximum entropy relative to the prior, subject to matching the first two moments of the posterior, gives updates in terms of derivatives of the log evidence,

$$\hat{w}_{t+1} = \hat{w}_t + C_t \frac{\partial \ln Z}{\partial \hat{w}}, \quad C_{t+1} = C_t + C_t \frac{\partial^2 \ln Z}{\partial \hat{w} \partial \hat{w}^\top} C_t, \quad (21)$$

and identically in the trust sector with $\mu_{e|r}, V_{e|r}$ in place of $\hat{w}, C$. Expressing the derivatives in the scaled fields yields (4)–(5) with the modulation functions (7). The first is a Hebbian term $\sigma_e \hat{x}$ with a tensorial, self-annealing rate; the rest is the online estimation of the channel.

G GENEALOGY OF THE LEARNING RULE

The update (4)–(5) descends from the variational optimization of modulated Hebbian on-line learning (Kinouchi & Caticha, 1992; Kinzel & Ruján, 1990), given a Bayesian reading by Oppen (1996; 1998) and applied across architectures by Solla & Winther (1999); the entropic-dynamics formulation we use is Caticha (2020). Learning through a noisy channel whose noise level must itself be inferred was analyzed by Kinouchi & Caticha (1993) and Biehl et al. (1995); Alves & Caticha (2016) identified that inferred level with distrust and made it dynamical, with applications to voting (Caticha et al., 2015) and to panels of appellate judges (Caticha & Alves, 2019), and a mean-field treatment by Simões & Caticha (2018). The blame-attribution behaviour resembles the least-action rule of Mitchison & Durbin (1989), which suffers retarded learning (Kabashima, 1994), whereas the optimized algorithm attains generalization error decaying as α^{-1} even under output noise (Simonetti & Caticha, 1996).

H THE PREJUDICE FIELD

Equation (8) is one filling of a 2×2 matrix D indexed by the classes of receiver and emitter, $h_w^p = h_w + D_{\kappa_r, \kappa_e}$. Four numbers are free. Since κ_r and κ_e each take two values, any filling decomposes uniquely into four orthogonal components,

$$D_{\kappa_r \kappa_e} = a + b \kappa_r + c \kappa_e + p \kappa_r \kappa_e, \quad (22)$$

with independent magnitudes and one meaning each. a shifts perceived agreement by the same amount whoever is speaking: a uniform credulity, referring to no label. b depends only on who is *listening*, so one class is systematically the more credulous. c depends only on who is *speaking*, so one class is believed more by everyone, its own members included. p depends only on whether the two classes *match*, which is in-group favouritism and out-group hostility at once, and is what (8) uses with $a = b = c = 0$.

Relabelling the classes sends $(a, b, c, p) \rightarrow (a, -b, -c, p)$. Only a and p survive it, and a does not refer to the label, so p is the one component that is both class-dependent and symmetric between the classes. That is why the main text uses it: under p alone the two classes are interchangeable, so any asymmetry in the outcome is spontaneous rather than imposed. The components that change sign, b and c , name a class rather than a relation between classes.

None of the four is redundant. The components are distinct because the modulation functions have a fixed scale. In particular, the uniform shift a moves the trust separatrix but not the opinion separatrix.

The four components pair with four ways trust can carry a label. Because trust is directed, its class dependence can be decomposed into the same four channels: uniform, receiver-specific, emitter-specific, and relational.

b and c are transposes, and nothing here can tell them apart. Receiver- and emitter-specific fields transpose the trust matrix. The symmetrized order parameters used in the main text cannot distinguish them.

What we have not run. The main text studies only p , with $a = b = c = 0$. We have not simulated the other components and make no claim about their collective phases.

I THE SIGN OF THE PREJUDICE FIELD

The sign of the prejudice field is fixed by the algorithm and not free. By (7), $F_\mu = (1 - 2\Phi(h_w))g(h_\mu)/Z$ is negative when $h_w > 0$, so perceived agreement drives distrust μ down. A receiver that adds $+p$ to h_w therefore becomes more trusting (tolerant) and one that adds $-p$ becomes more distrustful (intolerant). Equation (8) with $p > 0$ consequently gives in-group trust and out-group distrust, i.e. the discriminatory phase, which is the convention used throughout.

The companion manuscript (Anonymous, 2026) tabulates the in-group entry as $-p$ while its text and figures place the discriminatory phases at $p > 0$; those two statements are inconsistent, and taken literally the tabulated signs put the discriminatory phases at $p < 0$, mirroring every map in p . The same inconsistency appears in its description of the learning flow, where the panel labelled $D_{e|r} = -p$ is called the more tolerant one although $-p$ is the intolerant entry, and where the “distrust and disagree” basin is said to grow for $p > 0$ although adding $+p$ shrinks it. A single global sign reconciles everything; we adopt it in the form that leaves (8) untouched. The accompanying code implements both readings behind a flag, and Figure 14 runs the same sweep under each: the two rows are mirror images in p , and only the first matches the maps as published.

J CONVENTIONS IN THE ORDER PARAMETERS

Two choices in (12) deserve statement.

The class indicator. We use $G_{IJ} = \kappa_I \kappa_J \in \{-1, +1\}$. A $\{0, 1\}$ indicator, which counts only same-class pairs, cannot represent anti-correlation: under perfect reverse discrimination it returns $-(N/2 - 1)/(N - 1)$ rather than -1 , and its range depends on N . Since the reverse-discrimination half of the phase diagram is a substantive part of the result, the signed indicator is the appropriate one. Both are implemented.

Normalization. All three averages in (10) and (12) run over the $N(N - 1)/2$ unordered pairs, and the $\frac{1}{2}$ on the symmetrized trust puts all three in $[-1, 1]$. This matters because the three are read together as the colour channels of Figure 5; a literal reading of the model as previously written

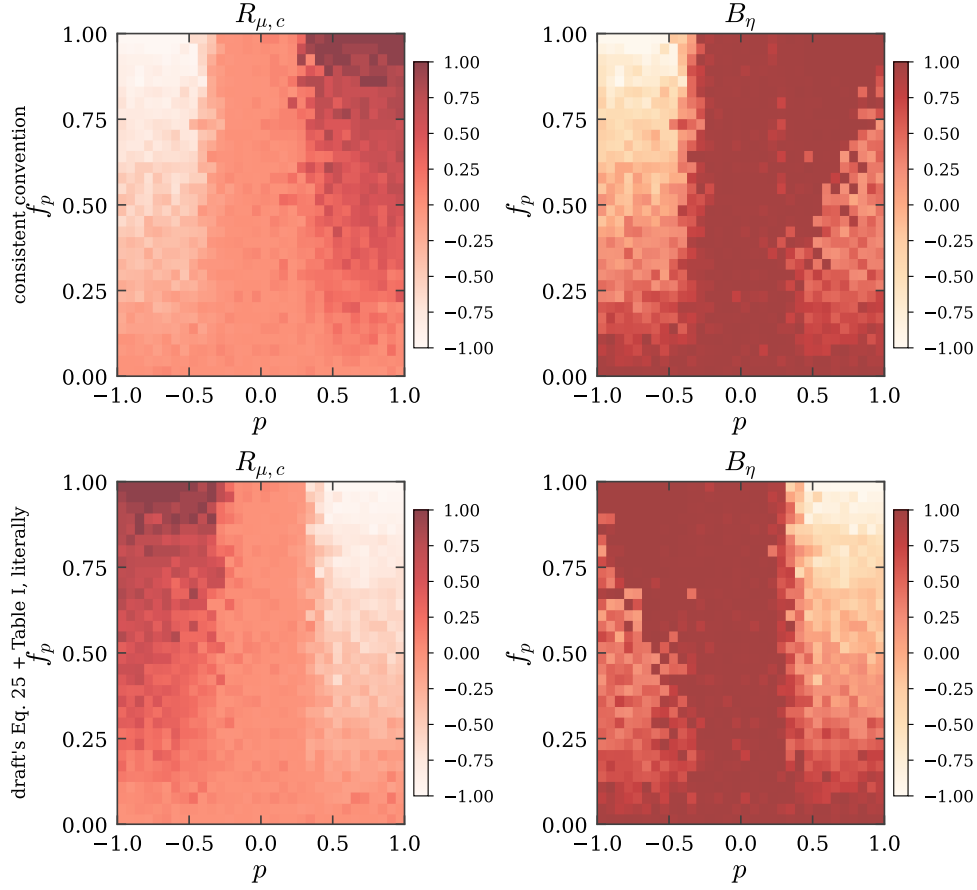


Figure 14: **The two readings of the prejudice field.** Trust–class correlation and trust balance over the (p, f_p) plane under the convention used in this paper (top) and under the companion manuscript’s table read literally with its Eq. 25 (bottom). The physics is identical; which half of the p axis is discriminatory is not.

normalizes $R_{c,w}$ by $N(N - 1)$ instead, capping it at $1/2$ and making the channels incommensurate. Both are implemented.

K PARAMETERS

The model needs six numbers. One is recovered from the companion manuscript, one is calibrated against its published figures, two are prior conventions, and two are free choices; all six are listed with their provenance rather than left implicit, since the manuscript states none of them. The accompanying code exposes each of them and its calibration script reproduces the comparisons below.

symbol	meaning	value	provenance
K	dimension of the issue embedding	30	recovered (see below)
N	agents in the society	40	chosen; converged (see below)
P	issues on the agenda	5 / 100	chosen: $\alpha = P/K$ either side of 1
Δt	interactions per ordered pair	500	calibrated (see below)
C_0, V_0	initial uncertainties	$I, 1$	uninformative prior
μ_0	initial distrust	$\mathcal{U}(-1, 1)$	gives $B_p \approx B_\eta \approx 0$ at $t = 0$

K is recovered, not chosen. The companion manuscript reports balance trajectories at $\alpha \in \{0.03, 0.17, 0.23, 0.33, 0.50, 0.67, 1.67, 3.33, 333.33\}$. These are exactly $P/30$ for $P \in \{1, 5, 7, 10, 15, 20, 50, 100, 10^4\}$, which fixes $K = 30$.

Δt is calibrated against the published trajectories. The dynamics anneals rather than reaching a stationary state, so the measurement time matters and the companion manuscript does not state it (its text carries a literal placeholder in place of the value). We scan Δt and compare the endpoints of all nine balance trajectories against the published ones:

Δt (interactions per ordered pair)	60	125	250	500	1000
rms distance to published endpoints	0.63	0.31	0.15	0.13	0.15

The minimum is flat-bottomed and sits at $\Delta t = 500$, which we use throughout. Seven of the nine endpoints then agree to within about 0.1, and B_ρ matches the three largest agendas almost exactly, but no single Δt reproduces all nine: two middle curves ($\alpha = 0.23, 0.33$) sit above ours, and the two largest agendas have a lower published B_η than we obtain at any Δt that fits the rest. Either the society size differs from ours or those middle curves, which overlap within a few pixels in the printed figure, cannot be digitized reliably; we cannot distinguish the two. Dropping the ambiguous readings does not move the selected value. All the Δt -independent checks pass: the above/below-diagonal ordering with its crossover at $\alpha \approx 1.7$, the sharp sign flip of $R_{\mu,c}$ at $p \approx 0$, and the $R_{c,w}$ wedge confined to large p and large f_p .

$N = 40$ is large enough. Repeating five representative points of the plane at $N \in \{20, 30, 40, 60\}$ with twelve realizations each, the order parameters at $N = 60$ differ from those at $N = 40$ by at most 0.02 ($R_{\mu,c}$, $R_{c,w}$), 0.03 (B_ρ) and 0.05 (B_η), while $N = 20$ differs by up to 0.24. The maps are therefore converged in N at $N = 40$, and the cost of going further is steep: total work grows as N^3 .

Numerical safeguards. Three floors are applied, none of which the model prescribes: on the evidence Z , whose exact modulation functions diverge in the deep dissonance corners; on the trust variance V , which $F_V < 0$ drives monotonically towards zero; and on the covariance update, clipped to the boundary at which $C + a(C\hat{x})(C\hat{x})^\top$ ceases to be positive semi-definite. The last is the only one that can alter a trajectory, and it never triggered in the runs reported here.